EXTENSION 1

Stage 6

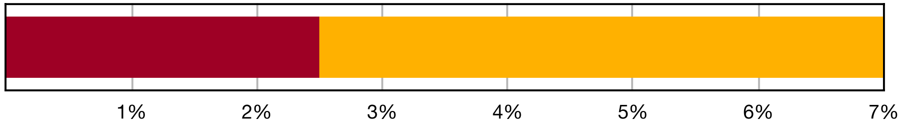
Trigonometry (Ext1), T3 Trig Equations (Y12)

Identities, Equations and 't' formulae (Ext1)

Teacher: SHS Maths

Exam Equivalent Time: 94.5 minutes *(based on allocation of 1.5 minutes per mark)*

T3 Trig Equations



Auxiliary Angles
Identities, Equations and 't' Formulae

***SmarterMaths analytics based on the average contribution to Extension 1 exams since 2020.**

HISTORICAL CONTRIBUTION

- *T3 Trig Equations* is easily the biggest sub-topic within Trigonometry and has contributed an impressive average of 7.0% per new syllabus Ext1 exam since its introduction in 2020.
- This topic has been split into two sub-categories for analysis purposes which are: *1-Auxiliary Angles (2.5%) and 2-Identities, Equations and 't' Formulae (4.5%).*
- This analysis look at *Identities, Equations and 't' Formulae.*

HSC ANALYSIS - What to expect and common pitfalls

- *Identities and Equations (4.5%)* was last examined in longer answer questions in 2023, 2021 (twice) and 2020. It was allocated a significant 4-marks within a very challenging 7-mark cross topic question in *2020 Ext1 14b*, a question that represents the highest level of difficulty that students can expect.
- A revision focus here is important. The use of compound and double angle identities is not only core knowledge for this topic area, but it will also prove extremely valuable in later applications of calculus.
- The 't' formulae was last examined in 2023 (we note its usage was not explicitly stated in the question) for the first time in 16 years. Ensure you revise and understand this solution methodology!
- NESA Topic Guidance explicitly states that students can be asked to provide "all" solutions to some trig equations. Carefully review *Ext1 T3 EQ-Bank 3* to see a worked solution that does not require knowledge of the general solution formulae.

Questions

1. Trigonometry, EXT1 T3 SM-Bank 10

Given that $\cot(2x) + \frac{1}{2}\tan(x) = a\cot(x)$, calculate a. *(3 marks)*

2. Trigonometry, EXT1 T3 EQ-Bank 6

Given $t = \tan\frac{x}{2}$, prove that $\frac{\sec x + \tan x}{\sec x - \tan x} = \left(\frac{1+t}{1-t}\right)^2$ *(2 marks)*

3. Calculus, EXT1 C2 2005 HSC 3b

i. By expanding the left-hand side, show that

$\sin(5x+4x) + \sin(5x-4x) = 2\sin(5x)\cos(4x)$ *(1 mark)*

ii. Hence find $\int \sin(5x)\cos(4x)\ dx$. *(2 marks)*

4. Trigonometry, EXT1 T3 2007 HSC 2a

By using the substitution $t = \tan\frac{\theta}{2}$, or otherwise, show that $\frac{1-\cos\theta}{\sin\theta} = \tan\frac{\theta}{2}$. *(2 marks)*

5. Trigonometry, EXT1 T3 2009 HSC 3c

i. Prove that $\tan^2\theta = \frac{1-\cos 2\theta}{1+\cos 2\theta}$ provided that $\cos 2\theta \neq -1$. *(2 marks)*

ii. Hence find the exact value of $\tan\frac{\pi}{8}$. *(1 mark)*

6. Trigonometry, EXT1 T3 2010 HSC 6a

i. Show that $\cos(A-B) = \cos A\cos B(1 + \tan A\tan B)$. *(1 mark)*

ii. Suppose that $0 < B < \frac{\pi}{2}$ and $B < A < \pi$.

Deduce that if $\tan A\tan B = -1$, then $A - B = \frac{\pi}{2}$. *(1 mark)*

7. Trigonometry, EXT1 T3 SM-Bank 13

Find all solutions of $\tan(2\theta) = -\tan\theta$ for $0 \leq \theta \leq 2\pi$. *(3 marks)*

8. Trigonometry, EXT1 T3 SM-Bank 11

Given that $\cos(\theta-\phi) = \frac{3}{5}$ and $\tan\theta\tan\phi = 2$, find $\cos(\theta+\phi)$. *(3 marks)*

9. Trigonometry, EXT1 T3 SM-Bank 12

Solve $\sin(2x) = \sin x$, for $0 \le x \le 2\pi$. *(3 marks)*

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10. Trigonometry, EXT1 T3 SM-Bank 8

Solve for x, given that

$x\sin(x)\sec(2x) = 0, \quad 0 \le x \le 2\pi$ *(2 marks)*

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11. Trigonometry, EXT1 T3 2021 HSC 11g

By factorising, or otherwise, solve $2\sin^3 x + 2\sin^2 x - \sin x - 1 = 0$ for $0 \le x \le 2\pi$. *(3 marks)*

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12. Trigonometry, EXT1 T3 2023 HSC 11e

Solve $\cos\theta + \sin\theta = 1$ for $0 \le \theta \le 2\pi$. *(3 marks)*

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13. Trigonometry, EXT1 T3 EQ-Bank 3

Find all values of θ that satisfy the equation $\sqrt{3}\cos\theta = \sin(2\theta)$. *(3 marks)*

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14. Trigonometry, EXT1 T3 EQ-Bank 5

By making the substitution $t = \tan\dfrac{\theta}{2}$, or otherwise, show that $\dfrac{1+\sin x - \cos x}{1 + \sin x + \cos x} = \tan\dfrac{x}{2}$. *(3 marks)*

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15. Trigonometry, EXT1 T3 SM-Bank 2

Show that

$\cos 3x = 4\cos^3 x - 3\cos x$. *(3 marks)*

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16. Trigonometry, EXT1 T2 2021 HSC 13d

i. The numbers A, B and C are related by the equations $A = B - d$ and $C = B + d$, where d is a constant.

Show that $\dfrac{\sin A + \sin C}{\cos A + \cos C} = \tan B$. *(2 marks)*

ii. Hence, or otherwise, solve $\dfrac{\sin\frac{5\theta}{7} + \sin\frac{6\theta}{7}}{\cos\frac{5\theta}{7} + \cos\frac{6\theta}{7}} = \sqrt{3}$ for $0 \le \theta \le 2\pi$. *(2 marks)*

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17. Trigonometry, EXT1 T3 EQ-Bank 1

i. Show that $\sin x + \sin 3x = 2\sin 2x \cos x$. *(2 marks)*

ii. Hence or otherwise, find all values of x that satisfy

$\sin x + \sin 2x + \sin 3x = 0, \quad x \in [0, 2\pi]$. *(2 marks)*

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18. Trigonometry, EXT1 T3 2005 HSC 4b

By making the substitution $t = \tan\dfrac{\theta}{2}$, or otherwise, show that

$\operatorname{cosec}\theta + \cot\theta = \cot\dfrac{\theta}{2}$. *(2 marks)*

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19. Trigonometry, EXT1 T3 2008 HSC 6b

It can be shown that $\sin 3\theta = 3\sin\theta - 4\sin^3\theta$ for all values of θ. (Do NOT prove this.)

Use this result to solve $\sin 3\theta + \sin 2\theta = \sin\theta$ for $0 \le \theta \le 2\pi$. *(3 marks)*

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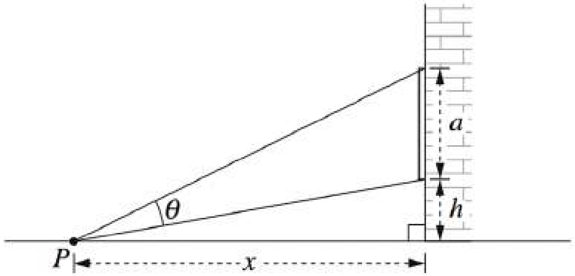
20. Trigonometry, EXT1 T3 SM-Bank 1

A billboard of height a metres is mounted on the side of a building, with its bottom edge h metres above street level. The billboard subtends an angle θ at the point P, x metres from the building.



Use the identity $\tan(A - B) = \dfrac{\tan A - \tan B}{1 + \tan A \tan B}$ to show that

$\theta = \tan^{-1}\left[\dfrac{ax}{x^2 + h(a+h)}\right]$. *(2 marks)*

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21. Trigonometry, EXT1 T3 2020 HSC 14b

i. Show that $\sin^3\theta - \dfrac{3}{4}\sin\theta + \dfrac{\sin(3\theta)}{4} = 0$. *(2 marks)*

ii. By letting $x = 4\sin\theta$ in the cubic equation $x^3 - 12x + 8 = 0$.

Show that $\sin(3\theta) = \dfrac{1}{2}$. *(2 marks)*

iii. Prove that $\sin^2\dfrac{\pi}{18} + \sin^2\dfrac{5\pi}{18} + \sin^2\dfrac{25\pi}{18} = \dfrac{3}{2}$. *(3 marks)*

Worked Solutions

1. Trigonometry, EXT1 T3 SM-Bank 10

$$\frac{1}{\tan(2x)}+\frac{\tan(x)}{2}=\frac{a}{\tan(x)},\quad \tan(x)\neq 0$$

$$\frac{1-\tan^2(x)}{2\tan(x)}+\frac{\tan(x)}{2}=\frac{a}{\tan(x)}$$

If $\tan(x)\neq 0$:

$$\begin{aligned}\frac{1-\tan^2(x)+\tan^2(x)}{2\tan(x)}&=\frac{a}{\tan(x)}\\ 1&=2a\\ \therefore a&=\frac{1}{2}\end{aligned}$$

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2. Trigonometry, EXT1 T3 EQ-Bank 6

$$\begin{aligned}\text{LHS}&=\frac{\sec x+\tan x}{\sec x-\tan x}\\ &=\frac{\frac{1+t^2}{1-t^2}+\frac{2t}{1-t^2}}{\frac{1+t^2}{1-t^2}-\frac{2t}{1-t^2}}\times\frac{1-t^2}{1-t^2}\\ &=\frac{1+t^2+2t}{1+t^2-2t}\\ &=\frac{(1+t)^2}{(1-t)^2}\\ &=\left(\frac{1-t}{1-t}\right)^2\end{aligned}$$

Worked Solutions

3. Calculus, EXT1 C2 2005 HSC 3b

i. $\sin(5x+4x)+\sin(5x-4x)=2\sin(5x)\cos(4x)$

$$\begin{aligned}\text{LHS}&=\sin(5x)\cos(4x)-\sin(4x)\cos(5x)+\sin(5x)\cos(4x)+\sin(4x)\cos(5x)\\ &=2\sin(5x)\cos(4x)\quad\ldots\text{ as required}\end{aligned}$$

ii. $\displaystyle\int\sin(5x)\cos(4x)\,dx$

$$\begin{aligned}&=\frac{1}{2}\int 2\sin(5x)\cos(4x)\,dx\\ &=\frac{1}{2}\int\sin(5x+4x)+\sin(5x-4x)\,dx\\ &=\frac{1}{2}\int\sin(9x)+\sin(x)\,dx\\ &=\frac{1}{2}\left[-\frac{1}{9}\cos(9x)-\cos(x)\right]+c\\ &=-\frac{1}{18}\cos(9x)-\frac{1}{2}\cos(x)+c\end{aligned}$$

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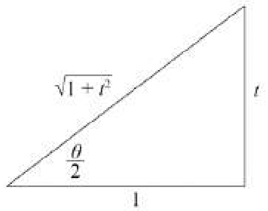
4. Trigonometry, EXT1 T3 2007 HSC 2a

COMMENT: The values of $\sin\theta$ and $\cos\theta$ can be committed to memory or quickly derived using double angle formulae (as shown in the worked solution).



$$\Rightarrow \tan\frac{\theta}{2} = t$$

$$\Rightarrow \sin\theta = 2\cdot\frac{t}{\sqrt{1+t^2}}\cdot\frac{1}{\sqrt{1+t^2}} = \frac{2t}{1+t^2}$$

$$\Rightarrow \cos\theta = \frac{1}{1+t^2} - \frac{t^2}{1+t^2} = \frac{1-t^2}{1+t^2}$$

Show $\dfrac{1-\cos\theta}{\sin\theta} = \tan\dfrac{\theta}{2}$:

$$\frac{1-\cos\theta}{\sin\theta} = \frac{1-\left(\frac{1-t^2}{1+t^2}\right)}{\frac{2t}{1+t^2}} \times \frac{1+t^2}{1+t^2}$$

$$= \frac{1+t^2-(1-t^2)}{2t}$$

$$= \frac{2t^2}{2t}$$

$$= t$$

$$= \tan\frac{\theta}{2} \quad \ldots\text{as required}$$

5. Trigonometry, EXT1 T3 2009 HSC 3c

i. Prove $\tan^2\theta = \dfrac{1-\cos2\theta}{1+\cos2\theta}$, $\cos2\theta \neq -1$

Using $\cos2\theta = 2\cos^2\theta - 1$

$$= 1-2\sin^2\theta$$

$$\text{RHS} = \frac{1-(1-2\sin^2\theta)}{1+(2\cos^2\theta-1)}$$

$$= \frac{2\sin^2\theta}{2\cos^2\theta}$$

$$= \tan^2\theta \ \ldots \text{as required.}$$

ii. $$\tan^2\frac{\pi}{8} = \frac{1-\cos\left(2\times\frac{\pi}{8}\right)}{1+\cos\left(2\times\frac{\pi}{8}\right)}$$

$$= \frac{1-\frac{1}{\sqrt2}}{1+\frac{1}{\sqrt2}} \times \frac{\sqrt2}{\sqrt2}$$

$$= \frac{\sqrt2-1}{\sqrt2+1} \times \frac{\sqrt2-1}{\sqrt2-1}$$

$$= \frac{\left(\sqrt2-1\right)^2}{2-1}$$

$$= \left(\sqrt2-1\right)^2$$

$$\therefore \tan\frac{\pi}{8} = \sqrt2 - 1 \qquad \left(\tan\left(\frac{\pi}{8}\right) > 0\right)$$

$(\tan\dfrac{\pi}{8} = \sqrt{3-2\sqrt2}$ is also a correct answer$)$

6. Trigonometry, EXT1 T3 2010 HSC 6a

i. Show $\cos(A-B)=\cos A\cos B(1+\tan A\tan B)$

$$\begin{aligned}\text{RHS}&=\cos A\cos B\left(1+\frac{\sin A\sin B}{\cos A\cos B}\right)\\&=\cos A\cos B+\sin A\sin B\\&=\cos(A-B)\ \ldots\text{as required}\end{aligned}$$

ii. Given $\tan A\tan B=-1$

$$\begin{aligned}\cos(A-B)&=\cos A\cos B(1-1)\\\cos(A-B)&=0\\A-B&=\cos^{-1}0\\&=\frac{\pi}{2},\frac{3\pi}{2},\ldots\end{aligned}$$

Since $0<B<\frac{\pi}{2}$ and $B<A<\pi$,

$$\Rightarrow A-B=\frac{\pi}{2}$$

7. Trigonometry, EXT1 T3 SM-Bank 13

$$\frac{2\tan\theta}{1-\tan^2\theta}=-\tan\theta$$

Let $\tan\theta=k$,

$$\begin{aligned}\frac{2k}{1-k^2}&=-k\\2k&=-k\left(1-k^2\right)\\2k+k\left(1-k^2\right)&=0\\3k-k^3&=0\\k\left(3-k^2\right)&=0\end{aligned}$$

$$k\left(3-k^2\right)=0\ \Rightarrow\ k=0,\ k=\pm\sqrt{3}$$

If $\tan\theta=0\ \Rightarrow\ \theta=0,\ \pi,\ 2\pi$

If $\tan\theta=\pm\sqrt{3}\Rightarrow\ \theta=\frac{\pi}{3},\ \frac{2\pi}{3},\ \frac{4\pi}{3},\ \frac{5\pi}{3}$

$$\therefore\theta=0,\ \frac{\pi}{3},\ \frac{2\pi}{3},\ \pi,\ \frac{4\pi}{3},\ \frac{5\pi}{3},\ 2\pi$$

8. Trigonometry, EXT1 T3 SM-Bank 11

$$\cos(\theta+\phi)=\cos\theta\cos\phi-\sin\theta\sin\phi$$

$$\cos\theta\cos\phi+\sin\theta\sin\phi=\frac{3}{5}\ \ldots(1)$$

$$\begin{aligned}\frac{\sin\theta\sin\phi}{\cos\theta\cos\phi}&=2\\\sin\theta\sin\phi&=2\cos\theta\cos\phi\ \ldots(2)\end{aligned}$$

Substitute (2) into (1):

$$\begin{aligned}\cos\theta\cos\phi+2\cos\theta\cos\phi&=\frac{3}{5}\\3\cos\theta\cos\phi&=\frac{3}{5}\\\cos\theta\cos\phi&=\frac{1}{5}\end{aligned}$$

Substitute $\cos\theta\cos\phi=\frac{1}{5}$ into (1):

$$\begin{aligned}\frac{1}{5}+\sin\theta\sin\phi&=\frac{3}{5}\\\sin\theta\sin\phi&=\frac{2}{5}\end{aligned}$$

$$\begin{aligned}\therefore\cos(\theta+\phi)&=\cos\theta\cos\phi-\sin\theta\sin\phi\\&=\frac{1}{5}-\frac{2}{5}\\&=-\frac{1}{5}\end{aligned}$$

9. Trigonometry, EXT1 T3 SM-Bank 12

$$\begin{aligned}2\sin x\cos x&=\sin x\\\sin x(2\cos x-1)&=0\end{aligned}$$

$$\sin x=0,\ \cos x=\frac{1}{2}$$

If $\sin x=0\ \Rightarrow\ x=0,\ \pi,\ 2\pi$

If $\cos x=\frac{1}{2}\ \Rightarrow\ x=\frac{\pi}{3},\ \frac{5\pi}{3}$

$$\therefore x=0,\ \frac{\pi}{3},\pi,\frac{5\pi}{3},2\pi$$

MARKER'S COMMENT: Many students expanded $\sin(2x)$ and then cancelled $\sin x$ on both sides which lost a set of solutions!

10. Trigonometry, EXT1 T3 SM-Bank 8

$$x\sin(x)\sec(2x) = \frac{x\sin(x)}{\cos(2x)} = 0$$

Find x that satisfies:
$x = 0$ and $\cos(2x) \neq 0 \Rightarrow x = 0$
or,
$\sin(x) = 0$ and $\cos(2x) \neq 0 \Rightarrow x = \pi,\ 2\pi$
$\therefore x = 0,\ \pi$ or 2π

11. Trigonometry, EXT1 T3 2021 HSC 11g

$$2\sin^3 x + 2\sin^2 x - \sin x - 1 = 0$$
$$2\sin^2 x(\sin x + 1) - (\sin x + 1) = 0$$
$$\left(2\sin^2 x - 1\right)(\sin x + 1) = 0$$

$2\sin^2 x = 1 \qquad \sin x = \ = -1$

$$\sin^2 x = \frac{1}{2}$$
$$\sin x = \pm\frac{1}{\sqrt{2}}$$

$$\therefore x = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{3\pi}{2}, \frac{7\pi}{4}$$

12. Trigonometry, EXT1 T3 2023 HSC 11e

Let $t = \tan\frac{\theta}{2}$

$$\frac{1-t^2}{1+t^2} + \frac{2t}{1+t^2} = 1 \text{ (see reference sheet)}$$
$$1 - t^2 + 2t = 1 + t^2$$
$$2t^2 - 2t = 0$$
$$2t(t-1) = 0$$
$$t = 0 \text{ or } 1$$

When $\tan\frac{\theta}{2} = 0$:
$$\frac{\theta}{2} = 0, \pi \Rightarrow \theta = 0 \text{ or } 2\pi$$

When $\tan\frac{\theta}{2} = 1$:
$$\frac{\theta}{2} = \frac{\pi}{4} \Rightarrow \theta = \frac{\pi}{2}$$

Test $\theta = \pi$:
$\cos\pi + \sin\pi = -1 + 0 = -1$ (not a solution)
$$\therefore \theta = 0, \frac{\pi}{2}, 2\pi$$

13. Trigonometry, EXT1 T3 EQ-Bank 3

$$\sqrt{3}\cos\theta = \sin(2\theta)$$
$$2\cos\theta\sin\theta - \sqrt{3}\cos\theta = 0$$
$$\cos\theta\left(\sin\theta - \frac{\sqrt{3}}{2}\right) = 0$$

$\cos\theta = 0$ or $\sin\theta = \frac{\sqrt{3}}{2}$
$\theta = 90^\circ, 270^\circ$ or $\theta = 60^\circ, 120^\circ \quad (0^\circ \leq \theta \leq 360^\circ)$

$\therefore \theta = 90^\circ + m \times 180^\circ,\ 60^\circ + m \times 360^\circ$ or $120^\circ + m \times 360^\circ$
where m is an arbitrary integer.

COMMENT: Expressing "all values" is specifically mentioned in Topic Guidance as an application of arithmetic sequences.

14. Trigonometry, EXT1 T3 EQ-Bank 5

$$\begin{aligned}\text{LHS} &= \frac{1+\sin x-\cos x}{1+\sin x+\cos x}\\ &= \frac{1+\dfrac{2t}{1+t^2}-\dfrac{1-t^2}{1+t^2}}{1+\dfrac{2t}{1+t^2}+\dfrac{1-t^2}{1+t^2}}\times\frac{1+t^2}{1+t^2}\\ &= \frac{1+t^2+2t-1+t^2}{1+t^2+2t+1-t^2}\\ &= \frac{2t(t+1)}{2(t+1)}\\ &= t\\ &= \tan\frac{x}{2}\end{aligned}$$

15. Trigonometry, EXT1 T3 SM-Bank 2

$$\begin{aligned}\text{LHS} &= \cos(2x+x)\\ &= \cos 2x\cos x - \sin 2x\sin x\\ &= \left(\cos^2 x-\sin^2 x\right)\cos x - 2\sin x\cos x\sin x\\ &= \cos^3 x-\sin^2 x\cos x-2\sin^2 x\cos x\\ &= \cos^3 x-\left(1-\cos^2 x\right)\cos x-2\left(1-\cos^2 x\right)\cos x\\ &= \cos^3 x-\cos x+\cos^3 x-2\cos x+2\cos^3 x\\ &= 4\cos^3 x-3\cos x\end{aligned}$$

COMMENT: High achieving students know the 3 variants of $\cos 2x$ back to front. Here, $\cos 2x = \cos^2 x - \sin^2 x$ breaks the back of this problem.

16. Trigonometry, EXT1 T2 2021 HSC 13d

i.

$$\begin{aligned}\frac{\sin A+\sin C}{\cos A+\cos C} &= \frac{\sin(B-d)+\sin(B+d)}{\cos(B-d)+\cos(B+d)}\\ &= \frac{2\sin B\cos d}{2\cos B\cos d}\\ &= \frac{\sin B}{\cos B}\\ &= \tan B\\ &= \text{RHS}\end{aligned}$$

ii. Let $A=\dfrac{5\theta}{7},\ C=\dfrac{6\theta}{7}$

♦ Mean mark 50%.

$$\begin{aligned}B &= \frac{A+C}{2}\\ &= \frac{1}{2}\left(\frac{5\theta}{7}+\frac{6\theta}{7}\right)\\ &= \frac{11\theta}{14}\end{aligned}$$

$$\begin{aligned}\tan\frac{11\theta}{14} &= \sqrt{3}\\ \frac{11\theta}{14} &= \frac{\pi}{3},\ \frac{4\pi}{3}\\ \therefore \theta &= \frac{14\pi}{33},\ \frac{56\pi}{33}\end{aligned}$$

17. Trigonometry, EXT1 T3 EQ-Bank 1

i. $\sin x + \sin 3x = \sin x + \sin 2x\cos x + \cos 2x\sin x$

$$\begin{aligned}
&= \sin x + 2\sin x\cos^2 x + \left(\cos^2 x - \sin^2 x\right)\sin x \\
&= \sin x + 2\sin x\cos^2 x + \cos^2 x\sin x - \sin^3 x \\
&= \sin x + 3\sin x\left(1 - \sin^2 x\right) - \sin^3 x \\
&= \sin x + 3\sin x - 3\sin^3 x - \sin^3 x \\
&= 4\sin x\left(1 - \sin^2 x\right) \\
&= 4\sin x\cos^2 x \\
&= 2\sin 2x\cos x \\
&= \text{ RHS}
\end{aligned}$$

ii. $\sin x + \sin 2x + \sin 3x = 0$

$$\sin 2x + 2\sin 2x\cos x = 0$$

$$\sin 2x(1 + 2\cos x) = 0$$

If $\sin 2x = 0$:

$$2x = 0, \pi, 2\pi, 3\pi, 4\pi$$

$$\therefore x = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}, 2\pi$$

If $\cos x = -\dfrac{1}{2}$:

$$\therefore x = \frac{2\pi}{3}, \frac{4\pi}{3}$$

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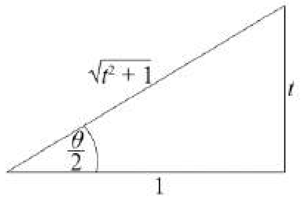
18. Trigonometry, EXT1 T3 2005 HSC 4b

Show $\operatorname{cosec}\,\theta + \cot\,\theta = \cot\,\dfrac{\theta}{2}$



$\tan\,\dfrac{\theta}{2} = t$

$$\Rightarrow \sin\theta = 2\sin\,\frac{\theta}{2}\,\cos\,\frac{\theta}{2} = \frac{2t}{t^2+1}$$

$$\Rightarrow \cos\theta = \cos^2\,\frac{\theta}{2} - \sin^2\,\frac{\theta}{2} = \frac{1-t^2}{t^2+1}$$

$$\Rightarrow \tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{2t}{1-t^2}$$

$$\begin{aligned}
\text{LHS} &= \frac{1}{\sin\theta} + \frac{1}{\tan\theta} \\
&= \frac{t^2+1}{2t} + \frac{1-t^2}{2t} \\
&= \frac{t^2+1+1-t^2}{2t} \\
&= \frac{1}{t} \\
&= \frac{1}{\tan\,\frac{\theta}{2}} \\
&= \cot\,\frac{\theta}{2} \\
&= \text{ RHS} \ \ldots \text{ as required.}
\end{aligned}$$

19. Trigonometry, EXT1 T3 2008 HSC 6b

Substitute $\sin 3\theta = 3\sin\theta - 4\sin^3\theta$
into $\sin 3\theta + \sin 2\theta = \sin\theta$,

$$\begin{aligned}
3\sin\theta - 4\sin^3\theta + \sin 2\theta &= \sin\theta \\
2\sin\theta - 4\sin^3\theta + 2\sin\theta\cos\theta &= 0 \\
2\sin\theta\left[1 - 2\sin^2\theta + \cos\theta\right] &= 0 \\
2\sin\theta\left[1 - 2\left(1 - \cos^2\theta\right) + \cos\theta\right] &= 0 \\
2\sin\theta\left[1 - 2 + 2\cos^2\theta + \cos\theta\right] &= 0 \\
2\sin\theta\left[2\cos^2\theta + \cos\theta - 1\right] &= 0 \\
2\sin\theta(2\cos\theta - 1)(\cos\theta + 1) &= 0
\end{aligned}$$

$2\sin\theta = 0$
$\Rightarrow \theta = 0, \pi, 2\pi$

$2\cos\theta - 1 = 0$

$\cos\theta = \dfrac{1}{2}$

$\Rightarrow \theta = \dfrac{\pi}{3}, \dfrac{5\pi}{3}$

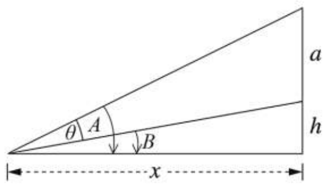
$\cos\theta + 1 = 0$

$\cos\theta = -1$

$\Rightarrow \theta = \pi$

$$\therefore\ \theta = 0, \frac{\pi}{3}, \pi, \frac{5\pi}{3}, 2\pi \quad (0 \le \theta \le 2\pi)$$

20. Trigonometry, EXT1 T3 SM-Bank 1



MARKER'S COMMENT: Answers that included a diagram and clearly labelled angles were generally successful.

Show $\theta = \tan^{-1}\left[\dfrac{ax}{x^2 + h(a+h)}\right]$

$\tan A = \dfrac{a+h}{x}$

$\tan B = \dfrac{h}{x}$

$$\begin{aligned}
\tan(A - B) &= \frac{\frac{a+h}{x} - \frac{h}{x}}{1 + \left(\frac{a+h}{x}\right)\left(\frac{h}{x}\right)} \times \frac{x^2}{x^2} \\
&= \frac{x(a+h) - xh}{x^2 + h(a+h)} \\
&= \frac{ax}{x^2 + h(a+h)}
\end{aligned}$$

Since $\theta = A - B$

$$\theta = \tan^{-1}\left[\frac{ax}{x^2 + h(a+h)}\right] \quad \ldots \text{as required.}$$

21. Trigonometry, EXT1 T3 2020 HSC 14b

i. Prove: $\sin^3\theta - \frac{3}{4}\sin\theta + \frac{\sin(3\theta)}{4} = 0$

$$\begin{aligned}
\text{LHS} &= \sin^3\theta - \frac{3}{4}\sin\theta + \frac{1}{4}(\sin2\theta\cos\theta + \cos2\theta\sin\theta) \\
&= \sin^3\theta - \frac{3}{4}\sin\theta + \frac{1}{4}\left(2\sin\theta\cos^2\theta + \sin\theta\left(1-2\sin^2\theta\right)\right) \\
&= \sin^3\theta - \frac{3}{4}\sin\theta + \frac{1}{4}\left(2\sin\theta\left(1 - \sin^2\theta\right) + \sin\theta - 2\sin^3\theta\right) \\
&= \sin^3\theta - \frac{3}{4}\sin\theta + \frac{1}{4}\left(2\sin\theta - 2\sin^3\theta + \sin\theta - 2\sin^3\theta\right) \\
&= \sin^3\theta - \frac{3}{4}\sin\theta + \frac{3}{4}\sin\theta - \sin^3\theta \\
&= 0
\end{aligned}$$

ii. Show $\sin(3\theta) = \frac{1}{2}$

Using part (i):

$$\frac{\sin(3\theta)}{4} = \frac{3}{4}\sin\theta - \sin^3\theta$$

$$\sin(3\theta) = 3\sin\theta - 4\sin^3\theta \ldots (1)$$

$$x^3 - 12x + 8 = 0$$

Let $x = 4\sin\theta$

$$\begin{aligned}
(4\sin\theta)^3 - 12(4\sin\theta) + 8 &= 0 \\
64\sin^3\theta - 48\sin\theta &= 0 \\
-16\underbrace{\left(3\sin\theta - 4\sin^2\theta\right)}_{\text{see (1) above}} &= -8 \\
-16\sin(3\theta) &= -8 \\
\sin(3\theta) &= \frac{1}{2}
\end{aligned}$$

♦♦♦ *Mean mark (iii) 21%.*

iii. Prove: $\sin^2\frac{\pi}{18} + \sin^2\frac{5\pi}{18} + \sin^2\frac{25\pi}{18} = \frac{3}{2}$

Solutions to $x^3 - 12x + 8 = 0$ are

$x = 4\sin\theta$ where $\sin(3\theta) = \frac{1}{2}$

When $\sin3\theta = \frac{1}{2}$,

$$3\theta = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{13\pi}{6}, \frac{17\pi}{6}, \frac{25\pi}{6}, \frac{29\pi}{6}, \ldots$$

$$\theta = \frac{\pi}{18}, \frac{5\pi}{18}, \frac{13\pi}{18}, \frac{17\pi}{18}, \frac{25\pi}{18}, \frac{29\pi}{18}, \ldots$$

$\therefore$ Solutions

$$x = 4\sin\frac{\pi}{18} \quad \left(= 4\sin\frac{17\pi}{18}\right)$$

$$x = 4\sin\frac{5\pi}{18} \quad \left(= 4\sin\frac{13\pi}{18}\right)$$

$$x = 4\sin\frac{25\pi}{18} \quad \left(= 4\sin\frac{29\pi}{18}\right)$$

If roots of $x^3 - 12x + 8 = 0$ are α, β, γ:

$$\alpha + \beta + \gamma = -\frac{b}{a} = 0$$

$$\alpha\beta + \beta\gamma + \alpha\gamma = \frac{c}{a} = -12$$

$$\left(4\sin\frac{\pi}{18}\right)^2 + \left(4\sin\frac{5\pi}{18}\right)^2 + \left(4\sin\frac{25\pi}{18}\right)^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \alpha\gamma)$$

$$16\left(\sin^2\frac{\pi}{18} + \sin^2\frac{5\pi}{18} + \sin^2\frac{25\pi}{18}\right) = 0 - 2(-12)$$

$$\therefore \sin^2\frac{\pi}{18} + \sin^2\frac{5\pi}{18} + \sin^2\frac{25\pi}{18} = \frac{24}{16} = \frac{3}{2}$$